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Numerical investigation of the influence of topography on simulated downburst wind fields

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ABSTRACT

A, dry, non-hydrostatic sub-cloud model is used to simulate an isolated stationary downburst wind event to study the influence topographic features have on the near-ground wind structure of these storms. It was generally found that storm maximum wind speeds could be increased by up to 30% because of the presence of a topographic feature at the location of maximum wind speeds. Comparing predicted velocity profile amplification with that of a steady flow impinging jet, similar results were found despite the simplifications made in the impinging jet model. Comparison of these amplification profiles with those found in the simulated boundary layer winds reveal reductions of up to 30% in the downburst cases. Downburst and boundary layer amplification profiles were shown to become more similar as the topographic feature height was reduced with respect to the outflow depth.

1. Introduction

The influence topographic features have on the near ground wind field during observed or simulated synoptic scale wind events are relatively well known and documented (e.g. Bowen and Lindley, 1977; Cao and Tamura, 2006; Glanville and Kwok, 1997; Holmes et al., 1997; Jackson and Hunt, 1975; Kim et al., 1997; Pearse et al., 1981; Taylor et al., 1987). This information is of great interest to those who design structures in areas of undulating or steep terrain, such as transmission towers or wind turbines. The influence of topographic features on the wind fields of much smaller mesoscale wind events, such as thunderstorm outflows, is however relatively unknown. The current level of knowledge is primarily based on simplified engineering outflow models using a steady flowing impinging air jet (Holmes, 1992; Letchford and Illidge, 1999; Mason et al., 2007; Selvam and Holmes, 1992; Wood et al., 2001), though the use of mesoscale numerical models has recently been investigated (Otsuka, 2006). These investigations have led to the general consensus that for isolated outflow events occurring in a still environment, topographic features locally accelerate winds, but the extent of acceleration is smaller than that found during large scale synoptic events. The primary reason for this difference has been linked to the unconfined nature of wall jet flows and the ability of the surface features to partially displace this relatively thin jet. Holmes (1992) however suggests that ambient wind conditions may influence the level of confinement, which in turn may influence acceleration levels. To the authors' knowledge there are currently no full-scale measurements of downburst wind fields measured above significant topographic features.

The aim of the current study is to use the axi-symmetric numerical sub-cloud downburst model described in Mason et al. (2009) to determine the speed-up, or amplification, effects of isolated 2D topographic features on the near surface wind velocities during simulated downburst events. The downburst model of Mason et al. (2009) uses an explicitly described thermal forcing function, as a parameterization of the true microphysics, to drive a downdraft within a non-hydrostatic, dry, sub-cloud domain. The model is axi-symmetric and therefore simulates a stationary (i.e. no simulated storm motion) event occurring in a still environment. General limitations as well as comparison with full-scale downburst observations are made in Mason et al. (2009); these limitations should be borne in mind when assessing results presented in this work. It should also generally be understood that the use of a dry model limits simulations to relatively idealised cases (Orf and Anderson, 1999), but is capable of producing model output similar to full-scale observations (Mason et al., 2009; Orf et al., 1996). The use

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of a numerical model allows the downburst problem to be simulated at full-scale and allows a study of the non-stationary characteristics of the wind field. The primary aim of this research was to determine whether the conclusions drawn with the simplified steady-state impinging jet models are reproduced when using a more complex, non-stationary simulation process. Results are presented describing the influence of both hills and escarpments for a range of upwind and downwind slopes and surface roughness. The influence of topographic feature size and location (with respect to the downdraft impingement point), as well as downdraft dimension, are also discussed.

2. Model description

A non-hydrostatic sub-cloud model, identical to that of Mason et al. (2009), was used in a dry, 2D axi-symmetric domain 6 km long and 4 km deep for this study. Model equations were solved in the anelastic form, and are:

$$\nabla \cdot (\rho_h \boldsymbol{U}) = 0 \tag{1}$$

$$\frac{\partial(\rho_h \boldsymbol{U})}{\partial t} + \nabla \cdot (\rho_h \boldsymbol{U} \otimes \boldsymbol{U}) = -\nabla p' + \nabla \cdot (\mu_{eff}(\nabla \boldsymbol{U} + \nabla \boldsymbol{U}^T)) + (\rho_h - \rho)\boldsymbol{g} \tag{2}$$

$$\frac{\partial(\rho_h \theta)}{\partial t} + \nabla \cdot (\rho_h \theta \boldsymbol{U}) = \nabla \cdot (\mu_{eff} \nabla \theta) + Q(x,y,z,t) \tag{3}$$

$$(\rho_h - \rho) = \frac{\rho_h}{\theta_h}(\theta - \theta_h) \tag{4}$$

where the subscript h is used to represent the reference atmospheric state, and Q is the parameterized forcing function. A neutral lapse rate was prescribed as the reference state for all simulations, and a 30 °C surface temperature led to a melting level of about 3 km. The model is closed using the SAS closure scheme (Menter and Egorov, 2005), which allows dynamic length scale calculations in unsteady regions of the flow for a better representation of turbulence characteristics (and the mean values based on them).

The model set of equations was shown in Mason et al. (2009) to produce a flow field similar to previously observed (Hjelmfelt, 1988) or simulated (Lin et al., 2007) downburst wind events for flow over a flat ground plane. To assess the ability of the model to handle flow over isolated terrain, simulations were performed for boundary layer winds over 2D bell-shaped hills and compared with wind tunnel experiments, Pearse et al. (1981). For a range of hill slopes, ϕ, Fig. 1 shows the model replicates mean wind tunnel topographic multipliers, M_t (Eq. (5)), to within 10% when measured above the hill crest (slightly higher for escarpments). Topographic multipliers and the hill slope variable, ϕ, are calculated using

$$M_t = \frac{U(z)_{topography}}{U(z)_{flat}} \tag{5}$$

$$\phi = \frac{H}{2 \cdot L_u} \tag{6}$$

where $U(z)_{topography}$ is the wind speed ($U=(u^2+w^2)^{1/2}$) at an elevation, z, above the local ground level, u and w are the horizontal and vertical wind speed components, respectively, and $U(z)_{flat}$ is the wind speed at the same elevation above the flat surface. H represents the topographic feature size (in this case 50 m), and L_u is the horizontal distance upwind from the hill crest to the location where the local ground elevation is $H/2$ above the flat surface level (Standards Australia, 2002). The numerical results shown in Fig. 1 were made using a similar grid and time step resolution to all downburst simulations (described in the next paragraph). Despite the similar setup though, the comparison shown in Fig. 1 can only be used as a general indication of the model capabilities when simulating downbursts. This is said because, to an unsteady Reynolds averaged Navier–Stokes (URANS) model, a constant boundary layer flow is treated as steady-state (although solved in an unsteady manner), and thus the SAS scheme closes the model equations primarily using the generic k–ε or k–ω equations (Menter and Egorov, 2005). For the highly unsteady case of the progressing downburst gust front however the scale adaptive component of the SAS model is invoked, thus a different set of closure equations are used for the transient gust front component. Despite this, and lack of any more representative comparative information, the relatively good performance of the model predictions above the crest shown in Fig. 1 gives some level of confidence that the model will handle the non-stationarity of the gust front acceptably in this region.

The downburst flow is resolved on a grid that inflates from 1 m at ground level to 20 m at an elevation of 2 km. A constant 5 m dimension is maintained radially to $x=3$ km from the downdraft centre. All simulations were performed with a time step of $\Delta t=0.25$ s. This grid and time step configuration led to a grid refined solution in the sense than an almost halving of grid size led to an increase in maximum velocity, U_{storm}, of only 2% for flow over the steepest hill slope tested ($\phi=1.0$); shallower slopes yielded smaller mesh induced variations. Model runs were solved using the commercial solver ANSYS CFX11, and all equations constants were the default values described in ANSYS (2007). All boundary conditions were set identical to Mason et al. (2009)

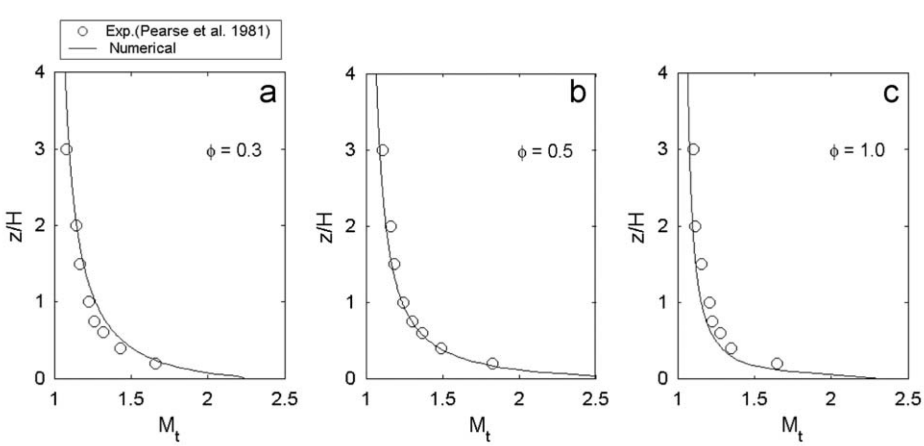

Fig. 1. Topographic multipliers predicted above the crest of 2D hills for simulated boundary layer flow over slopes of (a) ϕ=0.3, (b) ϕ=0.5, and (c) ϕ=1.0, and comparison with experimental wind tunnel measurements.

except the presence of a topographic feature on the ground plane. A z_0 value of 0.02 m was used for the ground plane roughness, except during the hill roughness tests where z_0 values of 0.002 and 0.2 m are also used.

Downburst simulations were performed for flow over both bell-shaped hills and escarpment features (Jackson and Hunt, 1975). The upwind shape for both topography types is identical, but while the downwind slope of the hill mirrors the upwind slope, the escarpment feature maintains a constant elevation equivalent to the crest height, H. The values of hill slope, ϕ, simulated were $0.1 \leq \phi \leq 1.0$; a range that allows simulation of both attached and separated downwind regions. The location of the topographic feature crest is at x_c=1.25 km corresponding to the location of maximum wind velocity, U_{storm}, in the baseline simulation over flat ground, Mason et al. (2009).

Mason et al. (2007) raise some issues relating to the implied three dimensionality an axi-symmetric model imposes on the shape of simulated topographic features (i.e. it essentially models circumferentially confined flow over an annulus). The same issues stand here, but subsequent three dimensional simulations with true two dimensional topographic features and three-dimensional downbursts have shown that differences in maximum wind speeds above feature crests are not significant within the high wind speed regions of flow. This agrees with the experimental findings of Mason et al. (2007). Therefore the findings presented are expected to reasonably represent true three-dimensional speed-up behaviour within the limitations of the simplified model implemented.

3. Results and discussion

3.1. Baseline simulation

The baseline run simulates the unsteady wind field of an isolated, stationary downburst wind storm over flat terrain. Each simulation begins with the forcing function removing energy from a prescribed region of the domain subsequently driving a downdraft of air. The resulting downdraft has a diameter of approximately 1 km, and maximum downdraft speed, $|w_{min}|$, of 36 m/s. As the downdraft moves towards the ground, the shear layer between the descending column of air and the quiescent surrounding environment rolls up into a ring vortex feature. As the downdraft and ring vortex impinge upon the surface, the flow is turned through 90° and diverges along the ground plane. Through the impingement process near surface winds accelerate to a maximum magnitude of $U_{storm} \approx 57$ m/s at an elevation of $z \approx 11$ m. As this peak is close to the surface, the direction of flow is essentially horizontal, thus the maximum horizontal velocity for the storm event, u_{storm}, is of the same magnitude. Fig. 2a shows the velocity structure ($0.1U_{storm}$ contours and U vectors) at the temporal point of U_{storm}, clearly showing the maximum wind speeds association with the region below the core of the diverging vortex. This finding is similar to full-scale (Hjelmfelt, 1988; Wilson et al., 1984) and laboratory scale observations (Alahyari and Longmire, 1995; Mason et al., 2005). As the front continues to diverge, a counter-rotating secondary vortex develops at the leading edge of the front between the high wind speed zone and the ground (i.e., at $x \approx 1500$ m in Fig. 2a). As the secondary vortex continues to develop, it acts as a lifting mechanism for the high wind speed region which is transported vertically between the upflow at the leading edge of the gust front and the secondary vortex lifted over it (Mason et al., 2009). As the vortex continues to diverge, it entrains ambient air and expands in size. This expansion combined with the vortex stretching process means velocities associated with the gust front drop-off rapidly. It is therefore the downdraft impingement that is critical for ultimate downburst design criteria

Viewing a sample of the domain in more detail, Fig. 2b shows the field of predicted maximum wind velocity, U, for the entire duration of the baseline event. Contours of U/U_{storm} are shown at

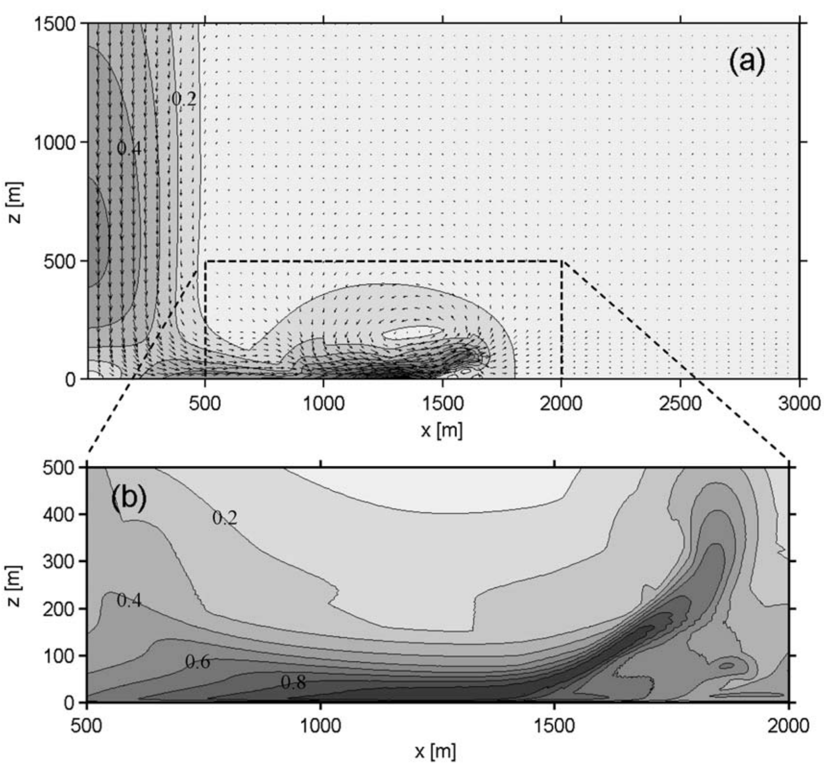

Fig. 2. Baseline simulation velocity structure, (a) at time of U_{storm} (t=338 s), velocity vectors and $0.1U_{storm}$ contours are shown, and (b) maximum U for the region 500 m $\leq x \leq$ 2000 m, 0 m $\leq z \leq$ 500 m, predicted to occur over the duration of the simulated downburst event.

0.1 intervals. Wind speeds within 10% of U_{storm} initially only influence the region $z < 50$ m, but as the secondary vortex is lifted over the leading edge of the front, these winds are lifted to $z \approx 150$ m. The importance the secondary vortex plays in the high wind speed structure of the event is evident. The presence of a secondary vortex in full-scale downburst flow is justified by Sherman (1987), but whether this vortex forms similarly in non-idealised situations is not known. Further discussion on this point is given in Mason et al. (2009).

3.2. Influence of topography shape

3.2.1. Flow structure

An example of simulated downburst wind structure as the burst front moves over a ϕ=0.6 hill is shown in Fig. 3. Wind speed, U, contours are again shown at intervals of $0.1U_{storm}$. Fig. 3b presents the structure at a similar time to the baseline simulation shown in Fig. 2a.

Fig. 3a shows that as the simulated front reaches the hill, winds accelerate in the vicinity of the crest and a recirculation vortex develops in the lee. As with boundary layer flow the extent of both acceleration above the crest and vortex development on the lee slope depend on upwind and downwind slope angles. For this example the maximum wind speed predicted above the crest is approximately 20% higher than that for flow over a flat ground surface. As the front moves over the hill, Fig. 3b, the wind speeds continue to intensify above the crest, and the lee vortex continues to develop in size. A high wind speed region has developed directly above the lee vortex which in effect appears to act as an extension to the topographic feature. The occurrence of lee slope vortices is well documented for boundary layer flow (e.g., Cao and Tamura, 2006; Kim et al., 1997; Pearse et al., 1981), but the unconfined nature of the isolated gust front appears to allow the vortex to grow to a larger size, at least temporarily, than expected in a developed (mean) boundary layer wind field. As the gust front continues to move over the hill the lee vortex is advected radially below the front, but collapses as the primary vortex moves past the hill itself, Fig. 3c. As the front continues to diverge from the hill, the maximum wind speed within the primary vortex decreases and the vortex expands in size. After the front has moved over the crest, the flow begins to stabilise in this region, and a steady-state regime, similar to that discussed in Mason et al. (2007) forms. Wind speeds above the crest for this steady flow region are of the order of 30–40% lower than those predicted to occur during the frontal passage. It is evident from this series of images that the simulated wind load felt by a structure at the top of the hill, although subject to wind speed that are extremely high, is loaded at that intensity for only a short duration of time.

3.2.2. Maximum velocity field

If the simulated wind field for the entire duration of storm event depicted in Fig. 3 is studied, it is possible to determine the maximum storm wind speed at any given point in the numerical domain, $U_{storm,x,y}$. As in Fig. 2b, this analysis allows an assessment of how the high wind speed regions are modified by the development of a lee-side or friction-induced counter-rotating

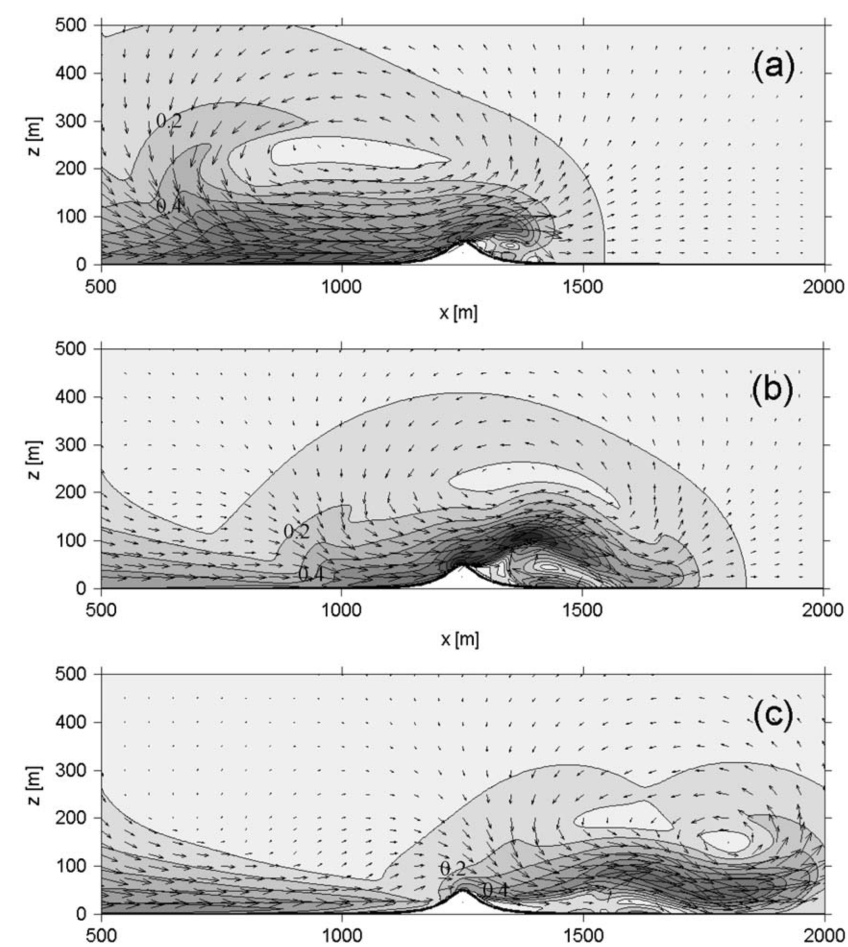

Fig. 3. Velocity vector and contour fields for times, (a) t=320 s, (b) t=340 s, and (c) t=360 s, for simulated downburst flow over a hill of slope, ϕ=0.6.

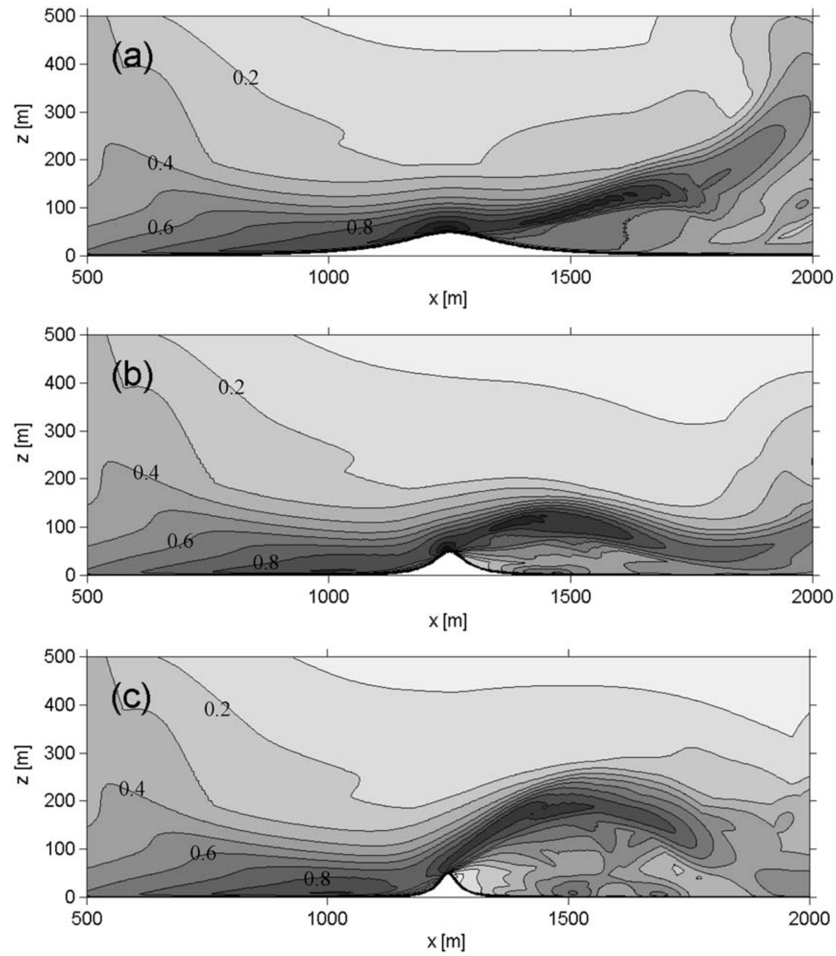

Fig. 4. Normalised maximum storm duration velocity, $U_{storm,x,y}/U_{storm}$, contour plots for simulated downburst flow over a: (a) shallow hill, $\phi=0.2$, (b) steep hill, $\phi=0.6$, and (c) very steep hill, $\phi=1.0$.

vortex throughout the life of a simulated storm. Fig. 4 plots the normalised $U_{storm,x,y}$ field for $\phi=0.6$, 0.2 and 1.0. This range of plots indicates the predicted flow fields over what could be considered a shallow, steep, and very steep slope, all of which can be viewed against the normalised $U_{storm,x,y}$ field for the flat ground plane, Fig. 2b. Again contours are normalised against U_{storm} and are plotted at intervals of $0.1U_{storm}$.

It is evident that the presence of a topographic feature significantly changes the appearance of the normalised $U_{storm,x,y}$ velocity field. Inspecting the field corresponding to $\phi=0.6$, Fig. 4b, a localised high wind speed region exists at the leading edge of, and just above the crest of the hill. This behaviour is as expected from any type of flow over a submerged obstacle. A secondary high wind speed region is seen at $x\approx1400$ m as the flow accelerates over the separation vortex behind the hill, Fig. 3b. This high wind speed region then continues to follow the shape of the vortex depicted in Fig. 3b and is associated with the movement of the primary vortex over the separation vortex prior to its decay, Fig. 3c. The normalised $U_{storm,x,y}$ field is seen to decelerate as it moves over the separation vortex, again as expected from the velocity fields shown in Figs. 3b and c. It is clear therefore that the maximum wind speeds for the entire event are still associated with the gust front region of the simulated downburst flow, but now the interaction with the topographic feature drives the exact location of maxima instead of the impingement/divergence process alone. Compared with the flow over a flat surface, one feature that is notably absent from the topography influenced flow field, Fig. 4b, is the rising high wind speed region associated with the interaction between the primary and secondary vortices. The absence of this wind region shows that the secondary vortex has not developed because the shearing forces required to grow this feature do not exist when the primary vortex is separated from the ground plane. Paradoxically this leads to the case where the presence of a topographic feature has lowered the elevation to which high wind speeds are predicted for the storm event. Taking the $0.1U_{storm}$ contour as a general indication of overall flow depth, comparing Figs. 2b and 4b, the elevation comparatively falls. In a non-idealised case less conducive to secondary vortex formation this however may not be the case.

The normalised flow field over a hill with $\phi=0.2$, Fig. 4a, shows a localised wind speed maximum over the crest of the hill with a secondary maximum further back in the rising velocity "tail". The rising high wind speed region in this case is something of an amalgam of both the previously discussed flow scenarios, i.e. the flat surface and the $\phi=0.6$ cases. In the case of Fig. 4a a counter-rotating vortex begins to form as the leading edge of the front moves over the crest of the hill, but instead of being overtopped as in Fig. 3c it is lifted over the front in a similar manner to the flat surface scenario. This lifting occurs because the flow quickly reattaches behind the developing counter-rotating vortex and then drives it forward and upward as in the flat surface simulation. The resulting flow field is thus made up of the flow acceleration at the crest of a topographic feature due to the fluid

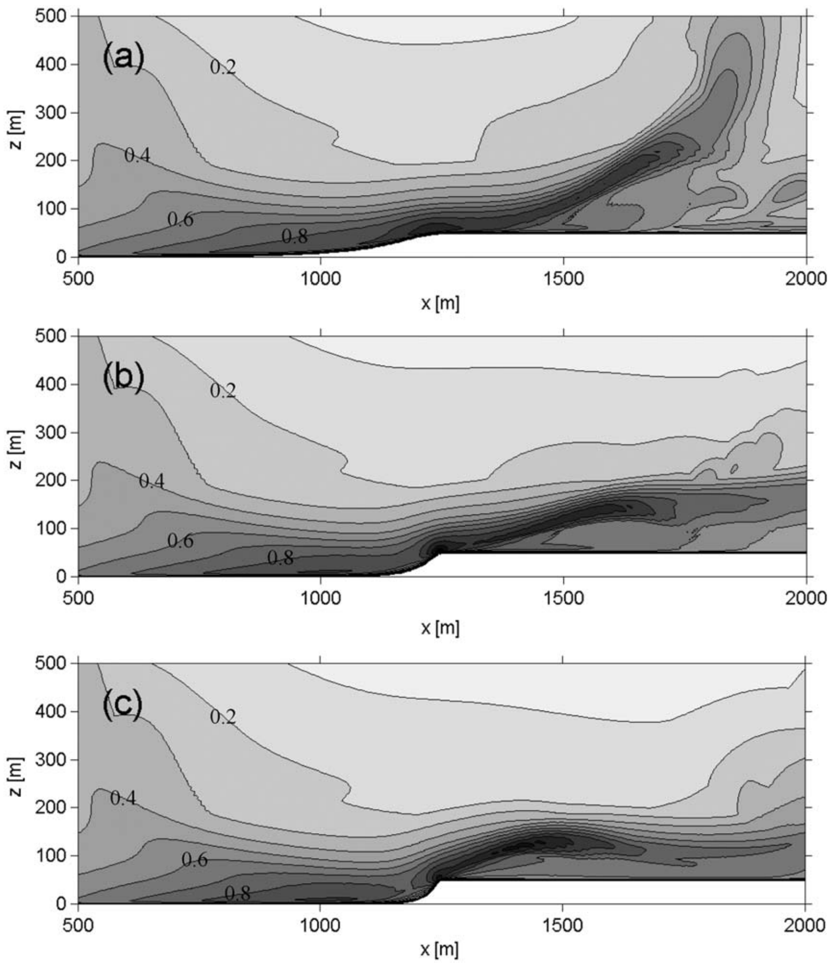

Fig. 5. Normalised maximum storm duration velocity, $U_{storm,x,y}/U_{storm}$, contour plots for simulated downburst flow over a: (a) shallow escarpment, ϕ=0.2, (b) steep escarpment, ϕ=0.6, and (c) very steep escarpment, ϕ=1.0.

essentially remaining attached over the hill crest, and the flow acceleration due to the interaction between the primary gust front vortex and the rising counter-rotating secondary vortex.

For flow over the very steep hill, ϕ=1.0, Fig. 4c, the same flow behaviour is seen as Fig. 4b, with the separation vortex dominating behaviour. However, the high wind speed region leaves the ϕ=1.0 hill at a greater angle than for the steep hill, leading to the formation of a larger lee-side vortex. A smaller overall value of velocity is therefore predicted because the flow is not distorted to the extent of other simulations.

The maximum wind speed characteristics for flow over escarpment features are shown in Fig. 5. As expected, localised accelerations again occur above the crest with an elevated tailing off zone. For an escarpment of ϕ=0.2, Fig. 5a shows an almost identical flow field to the flat surface case, Fig. 2b, except for the localised speed-up at the crest. The reason for such a replication is that the counter-rotating vortex formation process is predicted to be similar in both cases. Unlike for the shallow hill, the constant surface elevation after the crest allows the flow to remain attached and progress through the same shear-induced secondary vortex development process. However, for a slope of ϕ=0.6, Fig. 5b shows a velocity tail that turns downward; as was the case for the corresponding hill slope. This maximum wind field shape indicates the primary vortex no longer produces a shear-induced counter-rotating vortex, but instead develops a counter-rotating vortex due to flow separation as the front moves over the crest. The development method for this secondary vortex appears to be critical in the sense that when it is produced due to a shearing between the progressing front and the ground, the vortex forms at the front edge of the flow which then lifts it over the leading edge. However, when the counter-rotating vortex forms due to flow separation (and the flow is not re-attached within a short distance), the front will overtop, and subsequently destroy the secondary vortex as it diverges. This distinction alters the elevation at which high wind speeds are predicted and the shape of the maximum wind speeds observed in the lee of the crest. Predicted behaviour for flow over the escarpment of ϕ=1.0 are similar in shape and development process to ϕ=0.6, except for a small increase in elevation angle of the high wind speed region leaving the crest.

3.2.3. Maximum topographic multiplier

The elevation independent topographic multiplier, $M_{t,max}$, described in Mason et al. (2007), Eq. (7), can be used to reduce the maximum storm velocity information to a single value for each simulation. The aim of $M_{t,max}$ is to quantify the relative increase (or decrease) in the maximum predicted wind speed for each topographic simulation, $U_{storm,topography}$, by normalising it against the maximum wind speed predicted for flow over a flat surface, U_{storm}. Given maximum velocity is generally observed above the crest, $M_{t,max}$ gives an indication of the change in

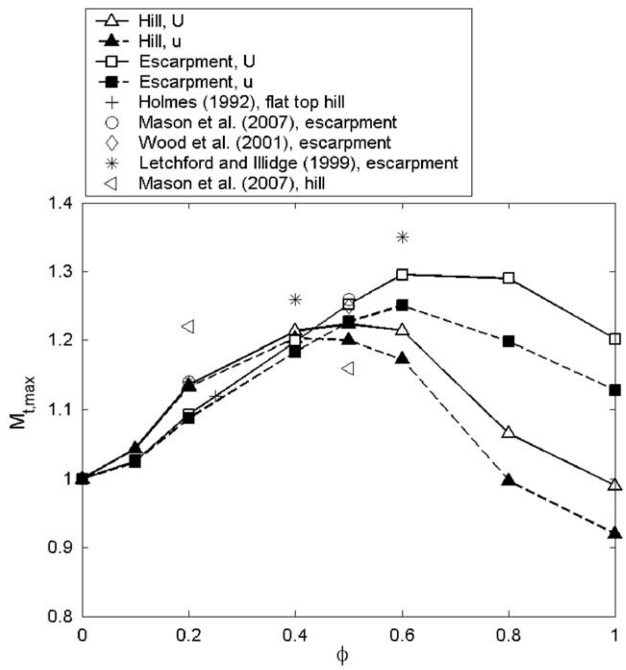

Fig. 6. Elevation independent maximum topographic multiplier, $M_{t,\ max}$, for flow over a range of hills and escarpments (Holmes, 1992, H/δ=0.25; Letchford and Illidge, 1999, H/δ=0.6; Mason et al., 2007 H/δ=0.35; Wood et al., 2001, H/δ=0.8).

maximum wind speed that a structure built near the crest is likely to be subjected to

$$M_{t,\max} = \frac{U_{storm,topography}}{U_{storm}} \tag{7}$$

Fig. 6 shows $M_{t,max}$ values predicted for all hill and escarpment features tested with the current simulation configuration. Mean experimentally measured values for steady flow impinging jet simulations are also shown (Holmes, 1992; Letchford and Illidge, 1999; Mason et al., 2007; Wood et al., 2001). $M_{t,max}$ values are shown for both U and u component velocities, both normalised against the corresponding flat surface U_{storm} or u_{storm} velocity. For flow over the hill $M_{t,max}$ values rise to a peak at ϕ=0.5 ($M_{t,max}$=1.22) for the U velocity, and ϕ=0.4 ($M_{t,max}$=1.20) for the u velocity, after which they decrease for increased topographic slope. The normalised U and u component $M_{t,max}$ values both follow a similar trend but the difference between the two values becoming larger for increased slopes. This behaviour is expected given the flow separates from the leading edge of the topographic feature at an angle primarily based on the slope of the topography. It is interesting to note that for very steep hills, i.e. ϕ=1.0, the maximum simulated wind speed is less than that predicted for flow over a flat surface. This appears to occur because the flow is redirected to such an extent, Fig. 4c, that little flow distortion occurs and the front leaves the hill at an angle similar to the slope itself. This indicates there is no rapid deformation of the flow and therefore little acceleration occurs.

For flow over the escarpment, the maximum $M_{t,max}$ value occurs at ϕ=0.6 for both U ($M_{t,max}$=1.30) and u ($M_{t,max}$=1.25) velocity components. For upwind slope angles $\phi \leq 0.4$ it is predicted that hills will accelerate winds more than escarpments with the same upwind slope, primarily because in these flow regimes the flow essentially remains attached through a larger deformation range because of the presence of a lee slope. For slope angles $\phi > 0.4$ however it is predicted that escarpments will produce larger velocity increases than similar shaped hills. The difference in this case is because the flow over the escarpments separate after it moves over the crest (Fig. 5) instead of just before the crest as is the case for hills (Fig. 4). This separation behaviour is also believed to be a primary reason why the maximum simulated $M_{t,max}$ value was predicted for an escarpment rather than for a hill.

Also plotted on Fig. 6 are mean $M_{t,max}$ values for previous experimental steady flow impinging jet simulations. Inspecting first the results of Mason et al. (2007) for flow over a hill it is seen that steady jet values are in the general range of the current simulation results, but they show a decrease in $M_{t,max}$ values from ϕ=0.2 to 0.5. There are insufficient data points to determine a clear trend in the experimental data, but it appears as though these points reach a maximum at a slope less than ϕ=0.5 predicted for the non-steady simulation. Inspecting the much larger set of experimental results for flow over escarpments, the general increasing trend for $\phi < 0.6$ in the current simulation data is again observed. The magnitude of all steady flow experimental results also appears to be within approximately 5% of the non-steady numerical simulations with a general tendency to be larger than these values. It is therefore evident that the amplification of the maximum wind speed within a simulation is similar in both steady and non-steady cases despite the flow fields being quite different. This is believed to occur because both simulations are based on a wall jet type flow that is unconfined at its upper boundary and is free to be displaced vertically by topographic features.

3.2.4. Velocity structure at the crest

It is pertinent from a design perspective to assess the wind structure in further detail at the general location of maximum intensity, the crest. Figs. 7 and 8 show U, (a), and u, (b), component velocities measured above the crest for the select simulations shown in Figs. 4 and 5 for hill and escarpment features, respectively. Velocity profiles shown are for the time of maximum storm velocity, $U_{storm\text{-}topography}$ or $u_{storm\text{-}topography}$, and are normalised against these values for flow over a flat surface (U_{storm} or u_{storm}). Normalised velocity profiles for simulated flow over a flat surface are also included for reference. Elevations are normalised against the topographic features height, H.

It is evident through inspection of Figs. 7 and 8 that for slopes $\phi \leq 0.6$ normalised U and u profiles are essentially identical in shape, with the u velocity profiles being marginally smaller in magnitude. This suggests that for flow over these slopes, in the near ground region, maximum winds are close to horizontal. For ϕ=1.0 however the flow structure is significantly different, particularly for $z/H < 1.0$. This difference, as discussed in Section 3.2.2, is due to the flow separating just prior to the crest giving the wind above the crest a significant vertical component. Figs. 7a and 8a also show that when the maximum U velocity predicted above the crest is larger than over a flat surface, i.e. $M_{t,max} > 1$, the elevation of occurrence decreases. This trend continues to the extent that the larger the $M_{t,max}$ value the smaller the elevation. Similar trends have been observed in experimental impinging jet flow (Mason et al., 2007; Wood et al., 2001). The opposite is also seen to be true, where in Fig. 7a the normalised U profile for flow over the hill of ϕ=1.0 shows a decrease in maximum predicted wind speed and an increase in the elevation of occurrence. This however is not seen to be the case for flow over the very steep escarpment. It is seen that the positive influence of topographic features on the maximum velocity profile is generally limited to the region $z/H < 1$ with the wind speeds above this generally being marginally lower than in the baseline case.

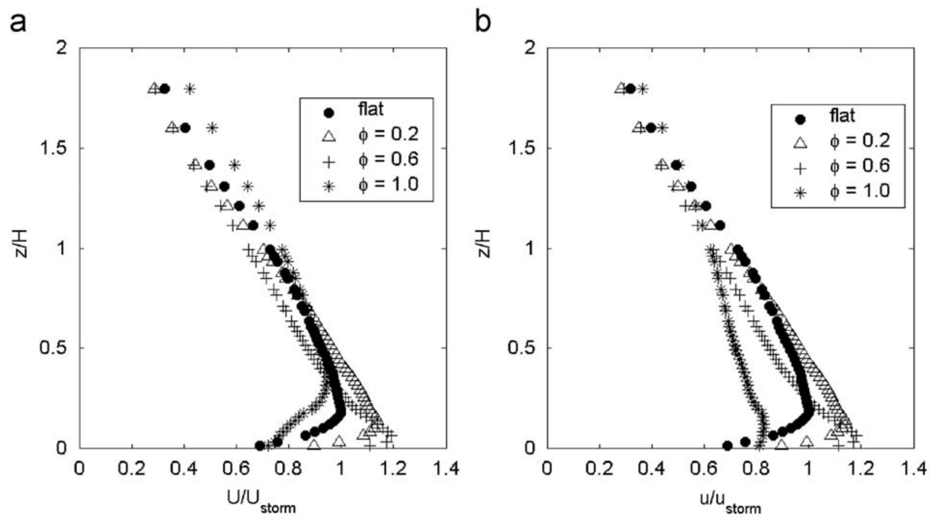

Fig. 7. Predicted normalised U, (a), and u, (b), velocity profiles above the crest for hills of variable slope at the time of $U_{storm\text{-}topography}$ or $u_{storm\text{-}topography}$.

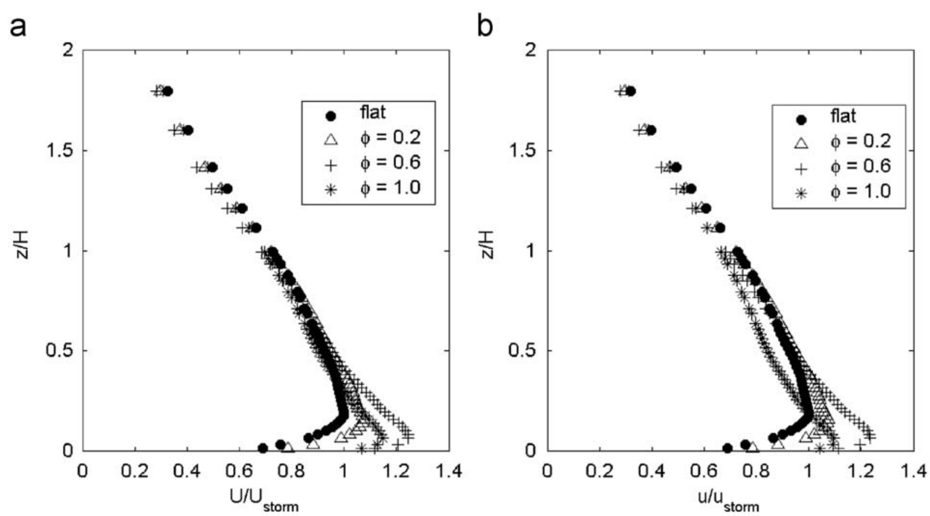

Fig. 8. Predicted normalised U, (a), and u, (b), velocity profiles above the crest for escarpments of variable slope at the time of $U_{storm\text{-}topography}$ or $u_{storm\text{-}topography}$.

3.2.5. Comparison with simulated boundary layer and steady flow impinging jet winds

It is common practice to reduce the influence of topography on boundary layer winds to a topographic multiplier, M_t. M_t profiles relate the wind speed at a specific elevation, z, above the local surface to the wind speed at the same elevation above a flat surface. Similar analysis is possible for the simulated downburst wind profiles discussed in Section 3.2.4, thus allowing direct comparison between the influences of topography on velocity profiles in the two flow regimes. Analysis of previous experimental steady flow impinging jet results is also possible. M_t is calculated using:

$$M_t = \frac{U(z)_{topography}}{U(z)_{flat}} \tag{8}$$

where the elevation, z, is taken as the elevation above the local ground level and both U velocities are taken at the time of maximum wind speed.

Fig. 9 shows M_t profiles for the simulated downburst flow over escarpments of slope ϕ=0.2 and 0.5. ϕ=0.5 is shown instead of ϕ=0.6 as used throughout the rest of the paper because there are more experimental data available for comparison at this slope angle. Fig. 9 shows clearly that the topography serves to increase the wind speed only below $z/H < 1$, above which point it is reduced. The extent of wind speed up, i.e. the magnitude of M_t, is greatest near ground level with maximum values of M_t=1.16 and 1.53 for slopes of 0.2 and 0.5, respectively. Given the velocity profiles shown in the previous section this form of the multiplier profile shape is expected because not only is the maximum wind speed above the crest increased due to the presence of topography, but the elevation at which it occurs is reduced. Similar profiles were observed for flow over all features except the very steep hill (ϕ=1.0), where $M_t < 1$ was observed for $z/H < 0.5$, and $M_t > 1$ for a limited region above this elevation. Similar M_t profiles are also observed for enveloped normalised location (enveloped velocity profile at a given location for the entire storm event) velocity profiles, suggesting that the profiles shown are representative of the velocity increase for event duration winds felt by a structure at a given location.

Comparing M_t profiles for the simulated downburst and boundary layer winds (Bowen and Lindley, 1977; Emeis et al., 1995) wind tunnel and full-scale, respectively), it is seen that at the top of the crest the simulated downburst velocity structure is accelerated to a lesser extent than the mean boundary layer velocities. For the two examples shown, there is up to a 30% reduction in velocity amplification when compared with the

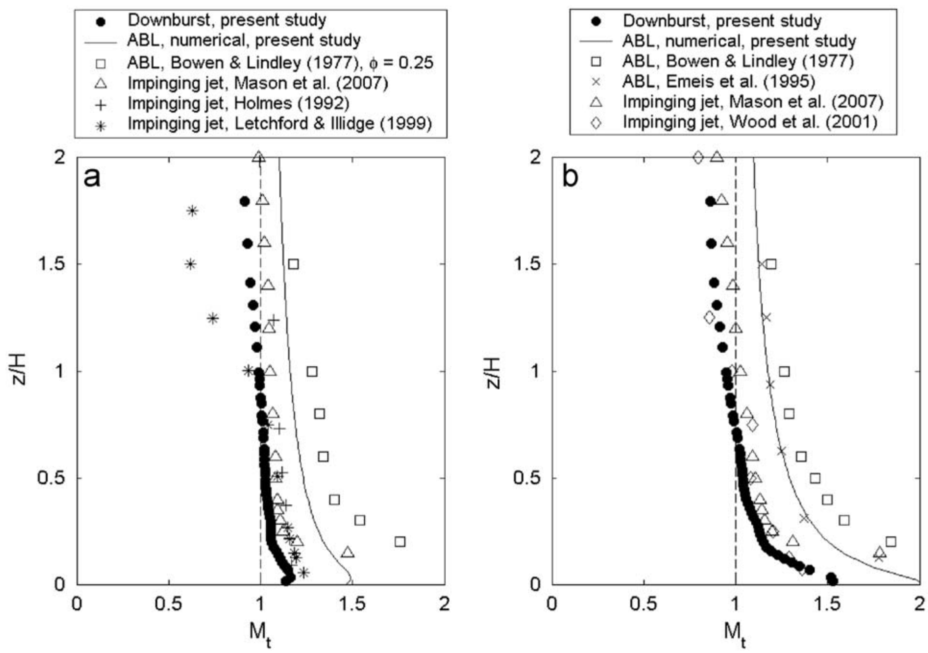

Fig. 9. Topographic multipliers for simulated downburst flow over an escarpment of: (a) ϕ=0.2, and (b) ϕ=0.5, as compared with boundary layer (Bowen and Lindley, 1977, wind tunnel H/δ=0.05; Emeis et al., 1995, full-scale H=16 m) and steady flow impinging jet downburst flow.

boundary layer experiments. Although the simulated downburst winds are not strictly "mean" values in the sense that the boundary layer measurements are, it is believed that this comparison is valid because the velocities used are based on the "moving-mean" values obtained numerically and do not include the turbulent component of the wind. If the turbulent components of both the flat surface and topographically influenced velocity profiles were included (and could be calculated reliably) perhaps a comparison with the smaller M_t values of the gust profile (Holmes, 2007) would be more appropriate. Comparisons for all other tested slopes provided similar results with the general decrease in M_t values reaching 30% in the regions of highest wind speeds.

Comparing current simulation results with those of experimental impinging jets, a good replication of profile shape is seen despite the two very different simulation methods. As with the $M_{t,max}$, the current simulations produce M_t values slightly lower than the steady impinging jet, but the replication of shape suggests the influence on bulk fluid behaviour is similar. This finding implies the velocity speed-up found in the current simulation is still governed largely by the unconfined nature of the upper bound of the wall jet, as suggested for impinging jet flow (Holmes, 1992). This finding also suggests that it may be possible to determine generic maximum wind speed amplification characteristics about topographic multipliers (above feature crests) by using a simplified impinging jet model. To determine temporal characteristics or lee region information however, a non-steady simulation must be run. A good replication of M_t profile shape was also found for flow over shallow hill features for which impinging jet simulation data were available (ϕ=0.2, Mason et al., 2007), but shapes began to differ slightly for flow over steeper hills (ϕ=0.5).

Table 1
$M_{t,max}$ values for the reference simulation cases, and all parametric tests discussed in Section 3.3.

	Escarpment ϕ=0.2	Escarpment ϕ=0.6	Hill ϕ=0.2	Hill ϕ=0.6
Reference	1.09	1.30	1.14	1.21
x_c=750 m	1.08	1.32	1.21	1.71
x_c=1750 m	1.11	1.35	1.17	1.41
H=25 m	1.11	1.24	1.18	1.12
H=100 m	1.05	1.15	1.08	1.14
D=3000 m	1.12	1.32	1.18	1.25
z_0=0.002 m	1.10	1.39	1.17	1.55
z_0=0.2 m	1.08	1.23	1.13	1.14

3.3. Parametric investigation

The discussion in Section 3.2 focused on the outflow of a single simulated downdraft. This section aims to determine the influence of changing simulation parameters on the outflow structure as the gust front moves over a topographic feature. The influence of crest location, x_c, and height, H, of the topographic feature are discussed for flow over ϕ=0.2 and 0.6 sloped isolated hills and escarpments. The influence of changing the downdraft diameter, D, and the simulated ground roughness, z_0, are also described. For each simulation the model parameters remain as for previous simulations, except for the model parameter in question. The only exception is the investigation of the influence of downdraft diameter where a larger numerical domain has been used (Mason et al., 2009).

All results in the current section are compared with those obtained in Section 3.2, referred to as the "Reference" results. Recapping, the reference results represent the simulated flow behaviour of a stationary D=1 km downburst diverging over an isolated topographic feature of a given slope, ϕ, positioned with a crest location of x_c=1250 m, a height of H=50 m, and an equivalent ground roughness of z_0=0.02 m. For ease of comparison, appropriate reference $M_{t,max}$ values are presented at this point along with results for all tests described in this section, Table 1. These values are introduced at this point so quick comparisons can be made for the bulk speed-up characteristics of each simulation, and to save repeating the reference results for each individual investigation. Throughout this section the flow structure over an escarpment of slope ϕ=0.6 is used to exemplify results, primarily because this flow combination produced the largest $M_{t,max}$ value in Fig. 6.

3.3.1. Influence of topography location

For the reference set of results the topographic feature was located at the position of maximum flat surface storm velocity, x_c=1250 m. To determine the influence of topographic location the topographic feature has been moved to two alternate positions, x_c=750 and 1750 m. The primary aim was to determine if changes in the gust front wind structure that occur during divergence influence the speed-up found in this region.

Fig. 10a shows the normalised maximum velocity profile predicted above the crest for the three escarpment locations. Velocity profiles for flow over a flat surface are shown for ready comparison. For both x_c=750 and 1750 m the presence of the escarpment is again predicted to increase wind speeds for $z/H < 1.0$ with a decrease in elevation of maximum winds also evident. Inspecting Fig. 10b, M_t profiles for each of the simulations show similar behaviour between tests, but show larger M_t values for $z/H < 0.2$ in the x_c=1750 m case. The increased values in this case occur because the elevation of the topographically influenced velocity peak is reduced from z/H=0.3 in the flat surface simulation to z/H=0.05 for flow over the escarpment. In comparison with the other locations the larger reduction in elevation means that relatively large velocities are normalised by relatively small velocities subsequently leading to large M_t values. It therefore appears that irrespective of the elevation of maximum velocity in the underlying velocity field, the presence of a topographic feature reduces the elevation of maximum velocity down to an elevation similar to those predicted in the reference simulations. Inspecting Table 1 however, even though the M_t profile in Fig. 10b varies between simulations, $M_{t,max}$ values are almost identical. This highlights that the location of the escarpment has little impact on the magnitude of maximum wind speed-up.

Viewing the $M_{t,max}$ variability for the shallow sloped, ϕ=0.2, escarpment and hill features, Table 1, similar repeatability between values is seen (all values within 5%). However, when inspecting $M_{t,max}$ values for the steep hill, ϕ=0.6, a 45% increase in $M_{t,max}$ is observed for the hill at x_c=750 m, and a 15% increase for x_c=1750 m when compared with the reference results. The reason for the difference at x_c=750 m is because of its fortuitous positioning at the point of descending ring vortex impact with the ground. This positioning means the flow over the crest is to a much larger extent confined (because of the negative w component of velocity), and flow over the crest stays attached to the surface longer. A similar prolonged attachment appears to occur for x_c=1750 m, but in this case is associated with the presence of the lifted counter-rotating vortex.

3.3.2. Influence of topography size

The logarithmic nature of boundary layer winds means that topographic multipliers (in a normalised z/H sense) are relatively independent of topography size except for a slight dependence on Jensen number (Holmes, 2001). However, the more complex and non-uniform structure of downburst wind fields mean this is not the case for these events. To investigate this further, downburst simulations have been run with topography heights of 25 and 100 m and compared with the reference simulation results over topography of 50 m.

Fig. 11 shows that distinctively different normalised profiles are present for each of the flat surface simulations. Viewing velocity profiles above the steep escarpment crest it is predicted that maximum winds will occur at approximately the same normalised elevation and intensity for flow over the 25 and 50 m features, however, the magnitude is significantly lower for flow over the 100 m escarpment. The same trend is evident in the $M_{t,max}$ values given in Table 1, but the discrepancy is smaller. The reason for the difference between what is seen in Fig. 11a and Table 1 is because the maximum velocity occurs just prior to the hill crest and is thus not evident in the plot. Despite this, it appears the reason for the reduction in maximum wind speed predicted above the largest escarpment is because the feature now significantly redirects the frontal propagation instead of simply distorting a smaller localised region as is the case for smaller features. Table 1 shows that similar behaviour occurs in each of the four simulations with H=100 m topography, but generally not to the extent depicted in Fig. 11a.

Inspecting the topographic multipliers in Fig. 11b the distinction between the two smaller and the largest feature is clearly evident. Again it should be noted that the flow in the 100 m simulation separates from the hill just prior (~10 m) to the crest, so if velocities at this location were included there would be slightly larger M_t values in this region. Another point of note is that M_t profiles for the 25 and 50 m escarpments are now distinctly separated. The movement of the H=25 m profile

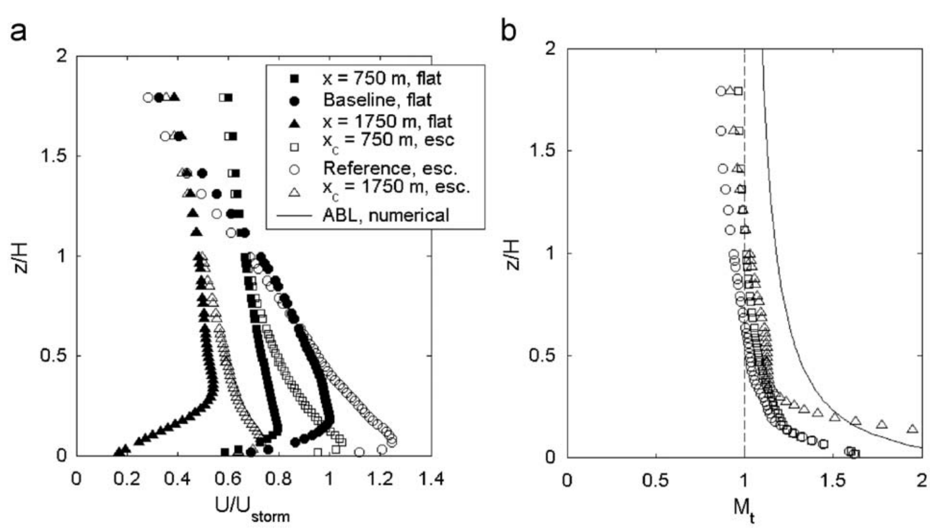

Fig. 10. (a) Velocity, and (b) topographic multiplier profiles for flow over escarpments positioned at x_c=750 m, x_c=1250 m (baseline), x_c=1750 m.

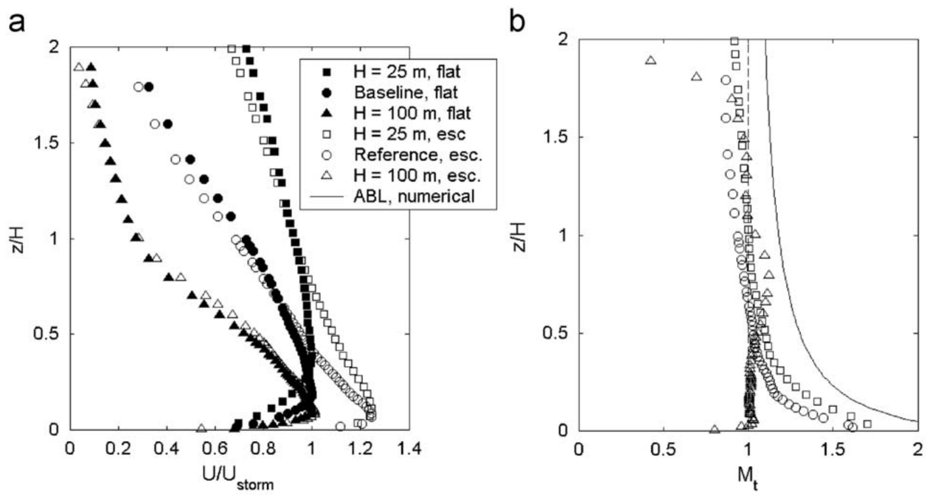

Fig. 11. (a) Velocity, and (b) topographic multiplier profiles for flow over escarpments of height, H=25, 50 (baseline) and, 100 m.

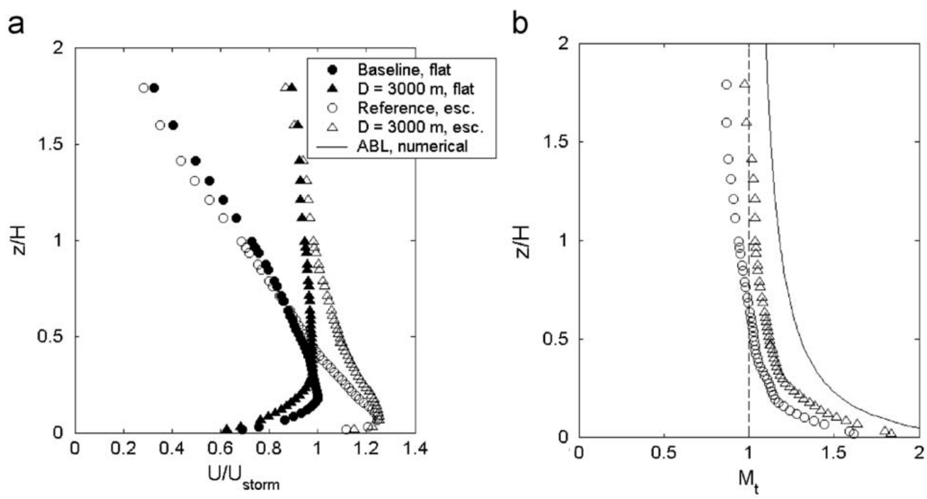

Fig. 12. (a) Velocity, and (b) topographic multiplier profiles for gust front flow over a ϕ=0.6 escarpment from downdrafts of D=1000 m (baseline) and D=3000 m.

towards the boundary layer profile suggests that the flow is now being influenced in a manner approaching that of the boundary layer. This trend is not surprising because as the simulated hill height decreases, the closer the hill height to flow thickness ratio gets to the boundary layer situation, and the further it moves from the reference case. This trend towards the boundary layer M_t profile was evident in all of the H=25 m simulations. Referring back to Table 1 reveals that this does not always lead to a larger $M_{t,max}$ value.

3.3.3. *Influence of downdraft size*

From field observations (Hjelmfelt, 1988; Fujita, 1985) it is evident that downbursts occur over a range of spatial scales from a few hundreds of metres to several tens of kilometres. Fujita (1985) suggests that of these, downbursts with damaging wind scales less than 4 km, microbursts, have the potential to produce the highest near ground wind speeds. Since it is these high wind speed events that are of importance for engineering design it is pertinent to investigate whether the physical size of a given downburst, within this range, will influence the speed-up effect on simulated winds. To investigate this simulations were run with a downdraft of diameter, D=3000 m for comparison with the reference events of D=1000 m. For the D=3000 m simulations the topographic feature crest was located at x=3000 m, which is again the location of maximum wind speeds in the flat surface simulation.

Inspecting Fig. 12a the increased depth of outflow and increase in elevation of maximum winds is clearly evident in the underlying flat surface velocity field. Inspecting the predicted escarpment influenced normalised velocity profile, the elevation of maximum velocity is reduced to $z/H \approx 0.05$, again irrespective of the initial position. The maximum velocity recorded above the escarpment crest is also seen to be approximately equal for the two cases leading to a small increase in $M_{t,max}$ for the larger downburst case as shown in Table 1. This small increase in $M_{t,max}$ when compared with the reference simulations is prevalent for all D=3000 m tests.

As was seen in Fig. 11b, Fig. 12b shows the increase in normalised flow depth shown in Fig. 12a leads to a shift in the M_t profile towards that of the boundary layer flow. The similarity between the M_t profiles of the D=3000 m and H=25 m simulations is expected given both simulations represent cases where the ratio of flow depth to topographic height is significantly larger than the reference tests.

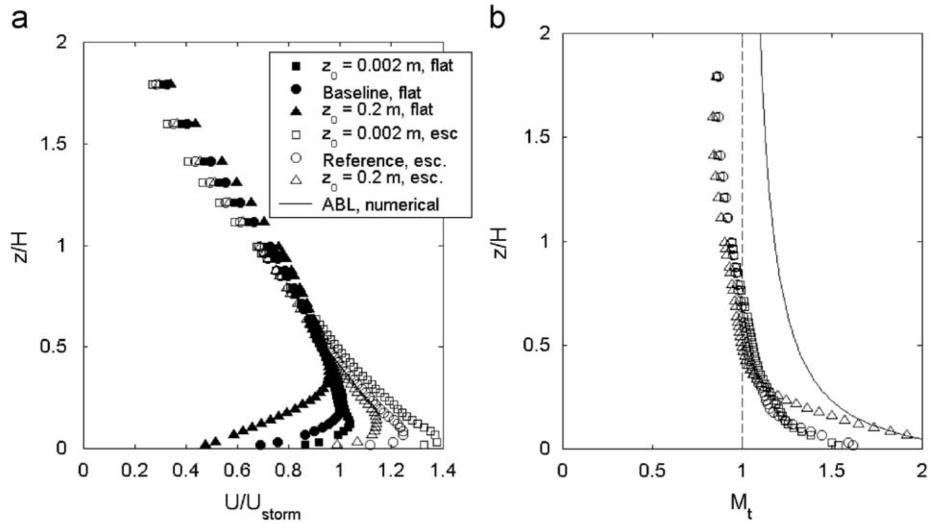

Fig. 13. (a) Velocity, and (b) topographic multiplier profiles for gust front flow over a ϕ=0.6 escarpments with surface roughness parameters of z_0=0.002, 0.02, and 0.2 m.

3.3.4. Influence of surface roughness

Surface roughness plays a significant role in determining boundary layer wind profiles and turbulence levels. It has been hypothesized that surface roughness should play less of a role in determining severe local storm profiles because of the reduced fetch over which it acts prior to reaching peak intensity, (Whittingham, 1964; Proctor, 1988) however suggests that ground friction can play an important role in determining simulated downburst structure and intensity. The latter finding was supported by Mason et al. (2009) and by the steady flow impinging jet simulations of Xu (2005). To investigate the role surface roughness plays in the induced amplification of winds over a topographic feature the reference tests were repeated for two additional surface roughness values, z_0=0.002 and 0.2 m. This section in essence measures the simulations dependence on the Jensen number (H/z_0).

For the simulated downburst flow over a flat surface an increase in the surface roughness parameter, z_0, leads to a predicted drop in normalised velocity magnitude and a rise in the predicted elevation of occurrence, Fig. 13a. The predicted velocity profiles over the escarpment continue to display these characteristics, but the elevation of maximum wind speeds again shift closer to the ground. Looking at the $M_{t,max}$ values in Table 1, decreasing values are seen for an increase in surface roughness. This decreasing trend is believed to occur because the rougher surfaces promote an earlier separation from the escarpment and thus less flow distortion in the region of the crest. For the shallower ϕ=0.2 cases (escarpment and hill), the range in $M_{t,max}$ values is only of the order of a few percent of the reference value, suggesting that in the presence of non-idealised roughness features this trend may be blurred. For the steeper ϕ=0.6 simulations however, the range in values for differing roughness categories are much larger, suggesting that this behaviour is unlikely to be limited to simulations.

Inspecting M_t profiles above the crest of the steep escarpment, Fig. 13b, for $z/H > 0.3$ there is little difference between the three z_0 values tested, but below this point the roughest simulation diverges from other predictions. This finding is again driven by the relocation of maximum wind speed to an elevation significantly lower than in the underlying flat surface profile. The increase in M_t for $z/H < 0.3$ was observed for all simulated events implying that the influence of surface roughness on wind speed-up is restricted to the very near surface region, a characteristic also observed by Cao and Tamura (2006) for boundary layer flow. The magnitude of M_t increase was found to be driven primarily by the position (if it occurred) of flow separation from the slope.

4. Conclusions

Stationary downburst wind events were simulated using an axi-symmetric, dry, non-hydrostatic sub-cloud numerical model. Despite their simplified nature simulated downbursts showed many characteristics of observed full-scale downbursts including a ring vortex at the leading edge of a diverging gust front.

The numerical downburst model was used to assess the influence idealised topographic features have on the near ground maximum velocity structure of the diverging front as it passes over both hill and escarpment features. As the front began to move over a topographic feature a counter-rotating vortex formed that was either overrun and subsequently destroyed by the front itself, or lifted over its face in a manner similar to the friction-induced secondary vortex described in Mason et al. (2009). It was generally found that if the frontal winds remained attached over the topographic feature the counter-rotating vortex behaved in the latter manner and led to high wind speed penetration to relatively large elevations.

For simulated wind events over both hill and escarpment features the storm maximum wind speeds may increase by up to 30% above the magnitudes predicted for flow over a flat surface. The elevation at which these storm maxima occur are also predicted to fall below those predicted for flow over a flat surface. Maximum topographic wind speed amplification, $M_{t,max}$, occurred for a hill slope of ϕ=0.5 and an escarpment slope of ϕ=0.6, with the escarpment value approximately 10% larger than that for the hill. Comparing the current $M_{t,max}$ predictions with those measured in steady flow impinging jet simulations a good replication of results was found. This replication implies that the steady-state impinging jet may be useful in determining bulk acceleration characteristics of downburst type flow.

When the current results were converted to topographic multipliers, M_t, for comparison with previous impinging jet simulations, a good replication of results was found. However, when comparing these M_t results with simulated boundary layer induced profiles, the velocity amplification was typically up to 20–30% less in the downburst maximum wind field. This finding supports the initial claim made by Holmes (1992) and supported by Letchford and Illidge (1999) and Mason et al. (2007), that stationary downburst winds are accelerated to a lesser extent than boundary layer winds by topographic features. The current simulations show that this behaviour is not limited to steady flow impinging jet simulations, but is evident also in transient gust front simulations that incorporate flow features similar to observed gust front vortices. The reasoning by Holmes (1992) that the reduction in wind speed-up for downburst type winds was due to the unconfined nature of the upper boundary of the wall jet also appears to be substantiated by the variable location of this upper bound in Figs. 4 and 5. Confinement of this upper bound by strong environmental winds may influence this finding.

A set of parametric tests were carried out to determine the relative influence on flow acceleration of topographic feature location and height, as well as the influence of downdraft diameter and surface roughness. Moving the topographic feature generally showed that the overall level of velocity amplification ($M_{t,max}$) was relatively unchanged for each of the positions tested, except if prolonged flow attachment occurred. These simulations also showed a clear relocation of maximum winds to the region $z/H < 0.1$, irrespective of their location in the flat surface simulations. This latter finding led to variability in M_t profiles for some feature locations. When comparing simulations with different size topographic features it was revealed that as the relative topographic size decreases flow acceleration begins to behave more like a boundary layer flow and a shift in the M_t profile occurred. This profile shift occurs because the flow depth to topographic height ratio begins to move closer to that of a boundary layer and thus the amplification effects in the flow shift accordingly. Similar conclusions were drawn from the large downdraft simulation test. Comparing flow over topographic features of variable surface roughness decreasing $M_{t,max}$ values were predicted for simulations with higher surface roughness values. This trend was linked to the rougher surfaces preventing flow from remaining attached over feature crests thus reducing the level of flow distortion in this region.

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